<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>7</td><td></td><td></td><td></td><td>Indicative content:

Set-up
Use the optical pin and cork to suspend the ruler from its mid-point.
Add plasticine to one end of the metre ruler so that it is balanced.
Use cotton loops to attach the weights to the ruler.

Taking measurements
Hang a 100 g mass / 1 N at one end of the ruler.
Hang a 200 g mass / 2 N on the other side.
Move it until balance is achieved.
Record the distances of the masses from the pivot.
Repeat for additional masses and distances.

Analysis
Calculate the clockwise and anticlockwise moment for each mass using the following equation (100 g = 1 N): moment = $F\ d$
Determine if the Principle of Moments is satisfied for each.

5–6 marks
Description of the apparatus set-up, method and analysis.
There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.

3–4 marks
Description of 2 out of 3 of the apparatus set-up, the method or the analysis or a limited description of all 3.
There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</td><td>6</td><td></td><td></td><td>6</td><td></td><td>6</td></tr>
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Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				1–2 marks Description of 1 out of 3 of the apparatus set-up, the method or the analysis **or** a limited description of 1 or 2 areas. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure*. *The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*						
				Question 7 total	**6**	**0**	**0**	**6**	**0**	**6**